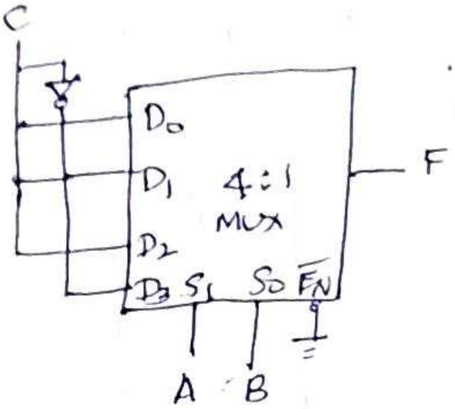
Method 2:

	$\bar{C}$	C	
C	0	1	D_0
C	2	3	D_1
C	4	5	D_2
$\bar{C}$	6	7	D_3

(a) Implementation table



(b) Multiplexer Implementation

* In the previous (method 1) multiplexer implementation two least significant variables, i.e., B and C are connected to the select lines of the multiplexer.
* The multiplexer implementation is also possible by connecting most significant variables i.e., A and B (in method2) to the select lines of the multiplexer.
* The procedure is also same which is explained below:
* Here, implementation table consists of two columns. The first ~~row~~ ~~column~~ column lists all the minterms where least significant variable is complemented ($\bar{C}$ in this case), and the second ~~row~~ column lists all the minterms with least significant variable is uncompleted (C in this case).
* The minterms given in the function are circled and then each row is inspected separately as follows:
  - If the two minterms in a row are not circled, 0 is applied to corresponding multiplexer input.

- If the two minterms in a row are circled, 1 is applied to corresponding multiplexer input.
- If the minterm in the column 1 is circled, least significant variable is complemented ($\bar{C}$ in this case) and applied to the corresponding multiplexer input.
- If the minterm in the column 2 is circled, least significant variable (C in this case) is applied to the corresponding multiplexer input.

1 Implement the following using 16x1, 8x1, 4x1 MUX.

$F(A, B, C, D) = \Sigma_m(1, 3, 4, 11, 12, 13, 14, 15)$

Decimal Value	A	B	C	D	F
0	0	0	0	0	0
1	0	0	0	1	1
2	0	0	1	0	0
3	0	0	1	1	1
4	0	1	0	0	1
5	0	1	0	1	0
6	0	1	1	0	0
7	0	1	1	1	0
8	1	0	0	0	0
9	1	0	0	1	0
10	1	0	1	0	0
11	1	0	1	1	1
12	1	1	0	0	1
13	1	1	0	1	1
14	1	1	1	0	1
15	1	1	1	1	1

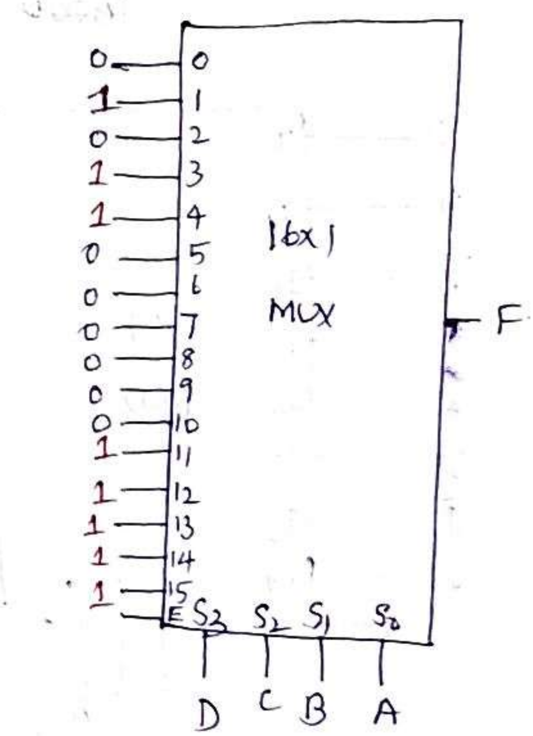
$A\bar{B}\bar{C}$, $\bar{A}B\bar{C}$, $A\bar{B}$, AB

<u>Case 1:</u> 16 x 1

$\therefore 2^4 = 16$

$n=4 \Rightarrow$ 4 select lines

Here, No. of select lines = No. of Variables



Case II: 8x1 MUX

8 to 1 MUX ⇒ $2^3 = 8$, $n = 3$ = no. of select lines = 3

no. of select lines < no. of variables of input

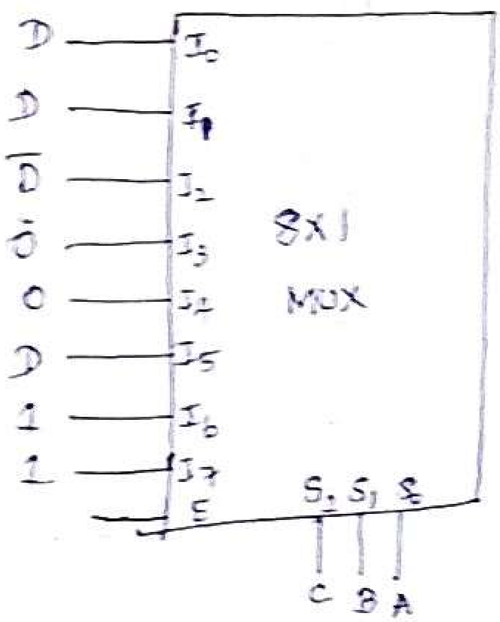
∴ Remaining 1 variable will be input variable and the other 3 will be selection variables.

(A, B, C → select lines, D → will be mapped as input)

	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
$\bar{D}$	0	2	5	6	7	8	9	10
D	1	3	4	11	12	13	14	15

	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
$\bar{D}$	0	2	4	6	8	10	12	14
D	1	3	5	7	9	11	13	15
	D	D	$\bar{D}$	0	0	D	1	1

$D + \bar{D} = 1$



Case 3: 4 X 1 MUX: ~~Case~~ : Method I:

4×1 MUX $= 2^{2} = 4$, $\overset{n}{\downarrow}$ 2 select lines

No. of select lines ≠ No. of variables of input

∴ 2 variables for select lines & 2 variables for input variables.

Here, A, B → Input variables

C, D → Select lines

	I_0	I_1	I_2	I_3
$\bar{A}\bar{B}$ (00)	0	1	2	3
$\bar{A}B$ (01)	4	5	6	7
$A\bar{B}$ (10)	8	9	10	11
AB (11)	12	13	14	15
	B	$A \odot B$	AB	$A+\bar{B}$

$I_3 \rightarrow \bar{A}\bar{B} + A\bar{B} + AB$

$\Rightarrow \bar{B}(A+\bar{A}) + AB$

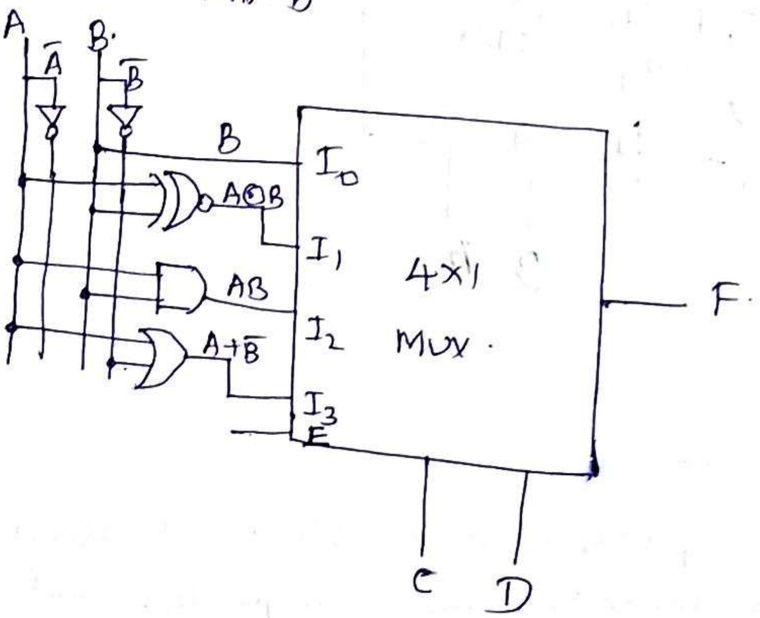
$\Rightarrow \bar{B} + AB$

$\Rightarrow \bar{B} + A$

$\boxed{\because A + \bar{A}B = A + B}$

$\Rightarrow \bar{A}B + AB$

$\Rightarrow B(A+\bar{A}) = B$



Method 2:

Here, C, D → Input variables

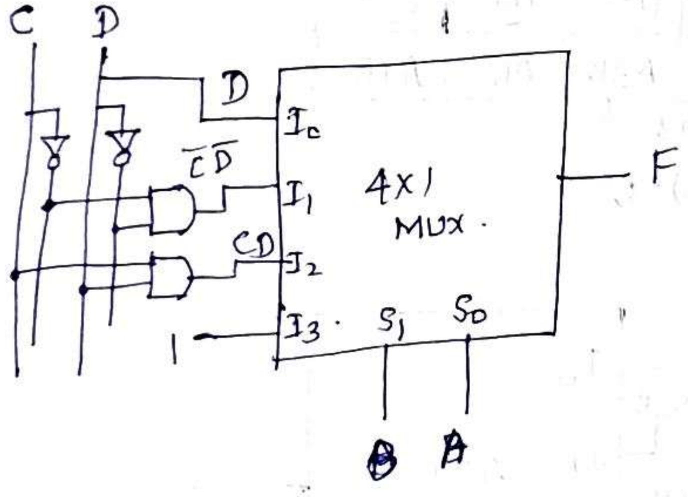
A, B → Select lines

	I_0	I_1	I_2	I_3
$\bar{C}\bar{D}$ (00)	0	4	8	12
$\bar{C}D$ (01)	1	5	9	13
$C\bar{D}$ (10)	2	6	10	14
CD (11)	3	7	11	15
	D	$\bar{C}\bar{D}$	CD	1

$I_0 = \bar{C}D + CD$

$= D(C+\bar{C}) = D$

$I_1 =$



2.2.9 Demultiplexer:

* A demultiplexer is a combinational circuit that receives information on a single line and transmits this information on one of 2^n possible output lines.

* The selection of specific output line is controlled by the values of n selection lines.

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* Fig. shows 1:4 demultiplexer.

Enable	S_1	S_0	D_{in}	Y_0	Y_1	Y_2	Y_3
0	X	X	X	0	0	0	0
1	0	0	0	0	0	0	0
1	0	0	1	1	0	0	0 ✓
1	0	1	0	0	0	0	0
1	0	1	1	0	1	0	0 ✓
1	1	0	0	0	0	0	0
1	1	0	1	0	0	1	0 ✓
1	1	1	0	0	0	0	0
1	1	1	1	0	0	0	1 ✓

Table: Function table for 1:4 demultiplexer

* The single input variable D_{in} has a path to all four outputs, but the input information is directed to only one of the output lines.

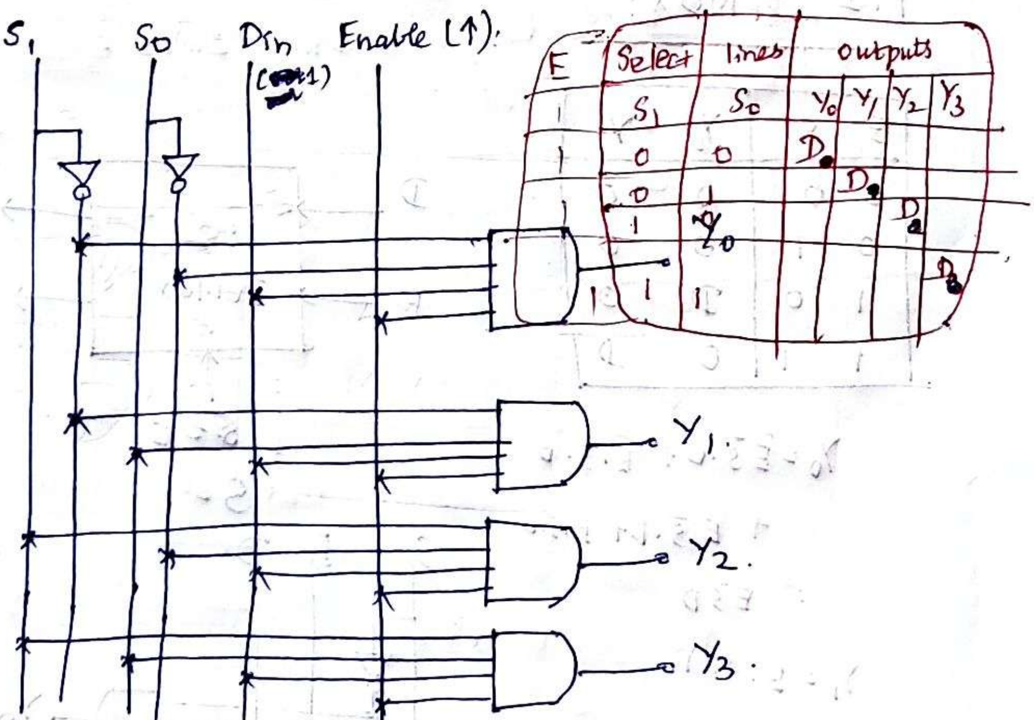
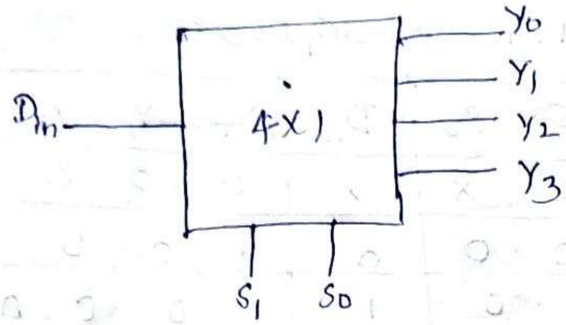


Fig: Logic diagram



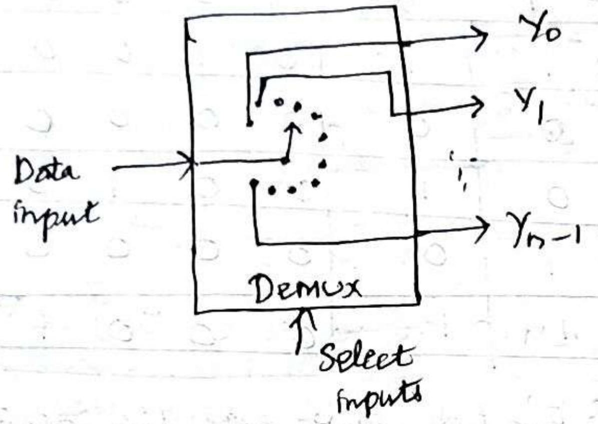
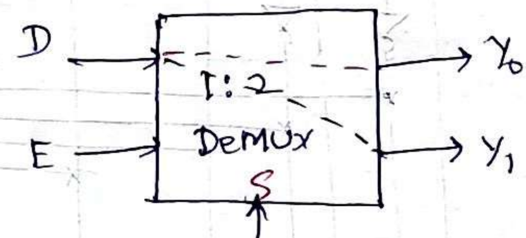


Fig: Functional diagram of a general demultiplexer

1:2 DeMUX:

E	S	Y_0	Y_1
0	0	0	0
0	1	0	0
1	0	D	0
1	1	0	D

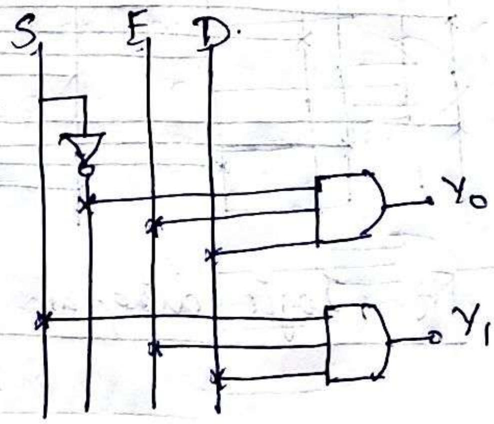


$S = 0$

$S = 1$

$$Y_0 = \bar{E}\bar{S}\cdot 0 + \bar{E}\cdot S\cdot 0 + E\bar{S}\cdot D + E\cdot S\cdot 0$$
$$= E\bar{S}D$$

$$Y_1 = E\cdot S\cdot D$$



Implementing Boolean function using DeMux:

1. $f(A,B,C) = \Sigma m(0,2,4,6)$ using 1:8 MUX.

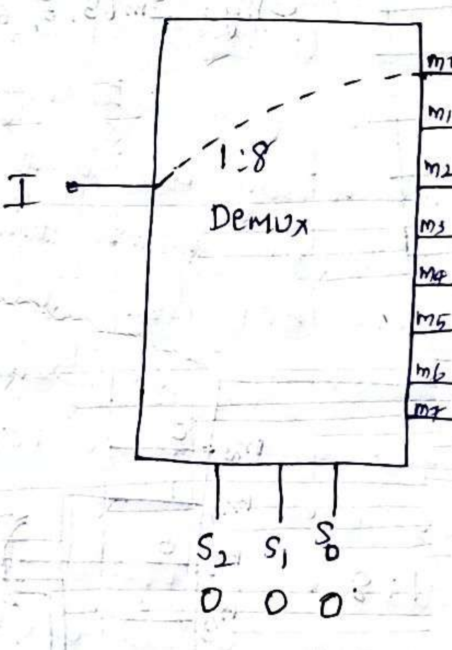
No of i/p variables $= 3 =$ No. of select lines

$1:2^n$

$1:2^3 = 1:8 \Rightarrow n = 3$. [1:8, 3 selection lines]

	S_2 A	S_1 B	S_0 C	Y f
0	0	0	0	1
1	0	0	1	0
2	0	1	0	1
3	0	1	1	0
4	1	0	0	1
5	1	0	1	0
6	1	1	0	1
7	1	1	1	0

S_2	S_1	S_0	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	0	0	0	I
0	0	1	0	0	0	0	0	0	I	0
0	1	0	0	0	0	0	0	I	0	0
0	1	1	0	0	0	0	I	0	0	0
1	0	0	0	0	0	I	0	0	0	0
1	0	1	0	0	I	0	0	0	0	0
1	1	0	0	I	0	0	0	0	0	0
1	1	1	I	0	0	0	0	0	0	0



$Y_0 = I = \bar{S_2}\bar{S_1}\bar{S_0} I = 000$

$Y_1 = 0 = \bar{S_2}\bar{S_1} S_0 I = 001$

$Y_2 = 0 = \bar{S_2} S_1 \bar{S_0} I = 010$

$Y_3 = 0$

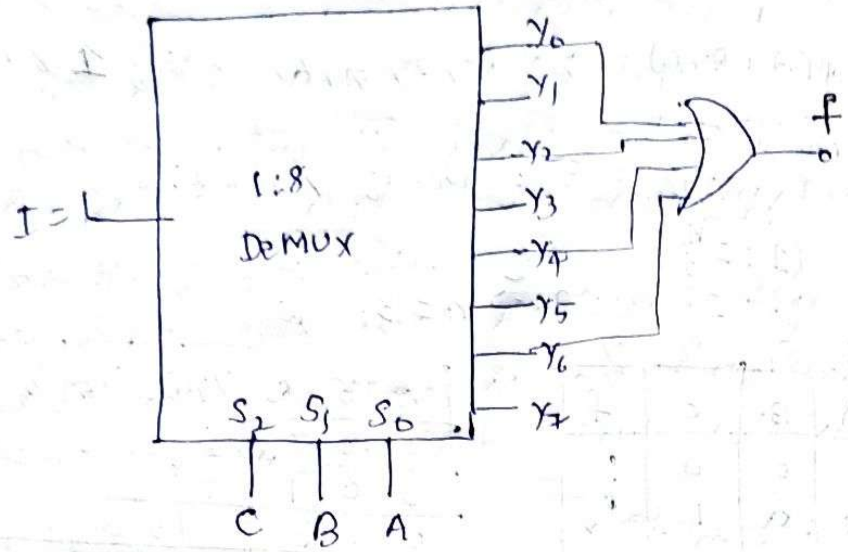
$Y_4 = 0$

$Y_5 = 0$

$Y_6 = 0$

$Y_7 = 0 = S_2 S_1 S_0 I = 011$

$f = \Sigma_m(0, 2, 4, 6)$



2. Implement full-adder using DeMux:

	A	B	C_{in}	Sum	Carry
0	0	0	0	0	0
1	0	0	1	1	0
2	0	1	0	1	0
3	0	1	1	0	1
4	1	0	0	1	0
5	1	0	1	0	1
6	1	1	0	0	1
7	1	1	1	1	1

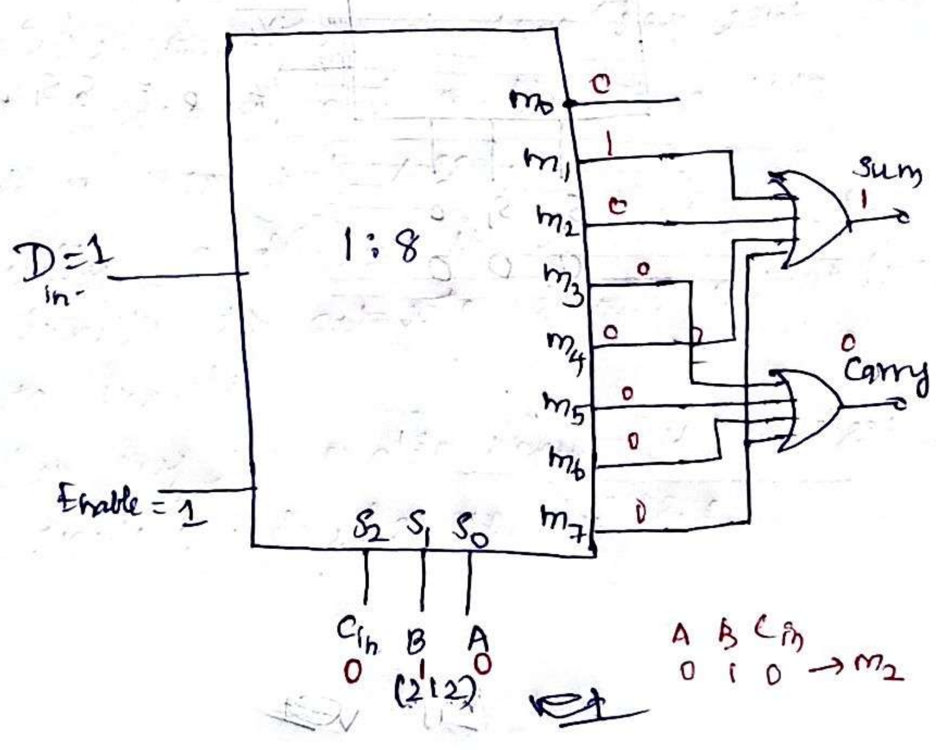
$1 : 2^n \Rightarrow 1 : 2^3 \Rightarrow n = 3.$

$1 : 8$

3 select lines

$Sum = \Sigma m(1, 2, 4, 7)$

$Carry = \Sigma m(3, 5, 6, 7)$

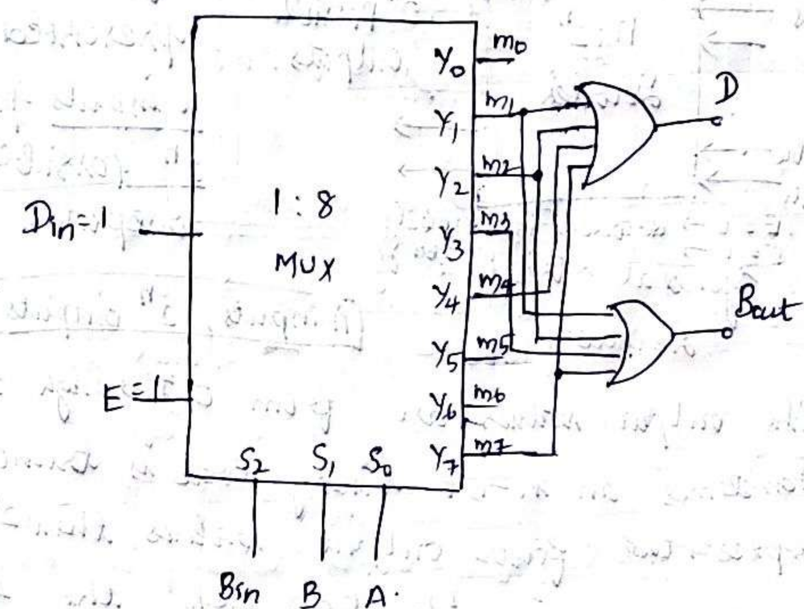


Implement full-subtractor using DeMux

A	B	B_{in}	D	B_{out}
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

$D = \Sigma m(1,2,4,7)$

$B_{out} = \Sigma m(1,2,3,7)$



2.2.10 Decoders:

* A decoder is a multiple-input, multiple-output logic circuit which converts coded inputs into coded outputs, where the input and output codes are different. The input code generally has few bits than the output code.

* Each input code word produces a different output code word, i.e., there is one-to-one mapping from input code words into output code words.

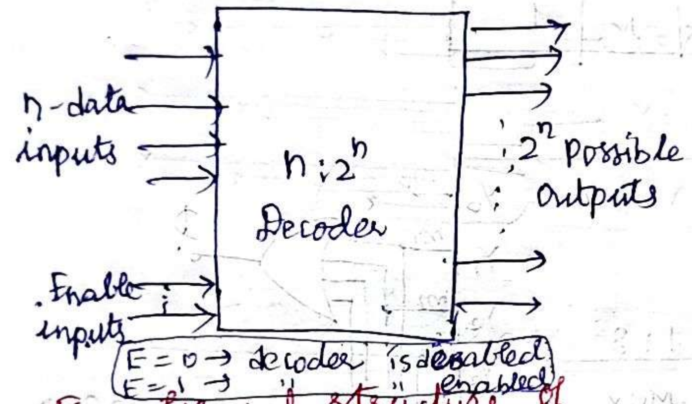


Fig: General structure of Decoder

* From the fig. the encoded information is presented as n inputs producing 2^n possible outputs.

n inputs, 2^n outputs with E=1

* The output values are from 0 through $2^n - 1$.

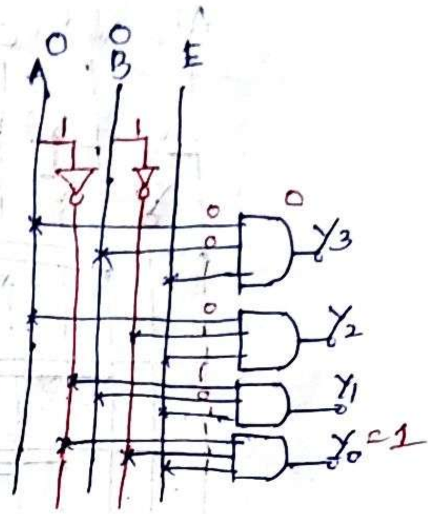
* Sometimes an n-bit binary code is truncated to represent fewer output values than 2^n.

* For example, in the BCD code, the 4-bit combinations 0000 through 1001 represent the decimal digits 0-9, and combinations 1010 through 1111 are not used.

* Usually, a decoder is provided with enable inputs to activate decoded output based on data inputs.

* When any one enable input is unasserted, all outputs of decoder are disabled.

2 to 4 decoder: (Binary decoder)



E	A	B	Y_3	Y_2	Y_1	Y_0
0	X	X	0	0	0	0
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0

$Y_3 = AB$, $Y_2 = A\overline{B}$, $Y_1 = \overline{A}B$, $Y_0 = \overline{A}\overline{B}$

3 to 8 decoder:

n inputs, 2^n outputs with E=1, no select lines

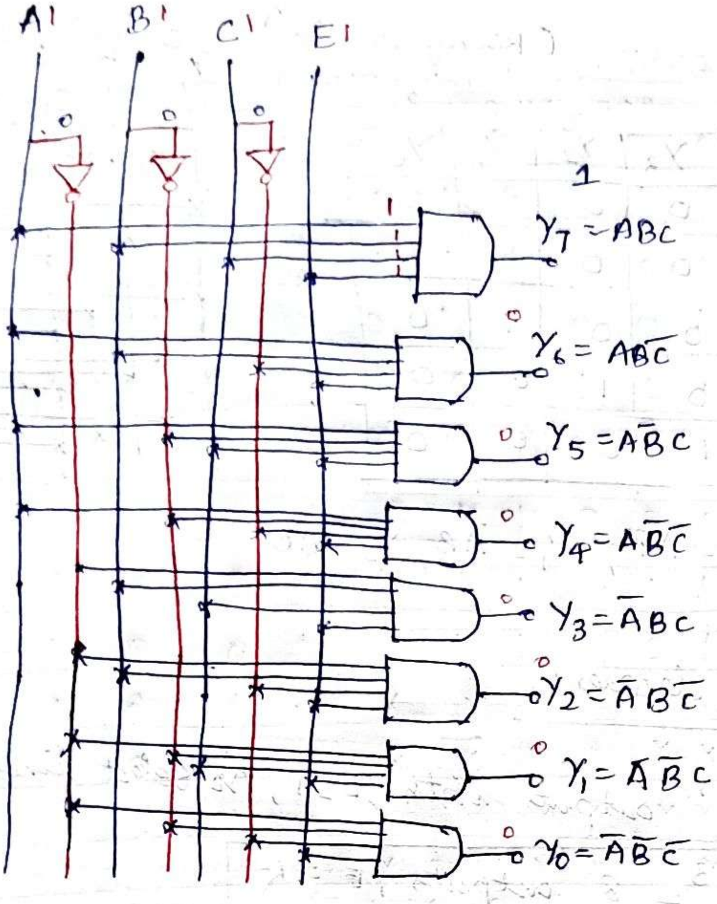
$\boxed{n=3,\ 2^{n=3} = 8 \text{ outputs}, E=1}$

E	A	B	C	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
0	X	X	X	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	1
1	0	0	1	0	0	0	0	0	0	1	0
1	0	1	0	0	0	0	0	0	1	0	0
1	0	1	1	0	0	0	0	1	0	0	0
1	1	0	0	0	0	0	1	0	0	0	0
1	1	0	1	0	0	1	0	0	0	0	0
1	1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	1	0	0	0	0	0	0	0

$Y_7 = ABC$, $Y_6 = AB\overline{C}$, $Y_5 = A\overline{B}C$, $Y_4 = A\overline{B}\overline{C}$

$Y_3 = \overline{A}BC$, $Y_2 = \overline{A}B\overline{C}$, $Y_1 = \overline{A}\overline{B}C$, $Y_0 = \overline{A}\overline{B}\overline{C}$

Binary decoder: A decoder which has an n-bit binary input code and a one activated output out of 2^n output code is called Binary decoder. It is used when it is necessary to activate exactly one of 2^n outputs based on an n-bit input value.



2.2.10.3 **BCD to 7-segment Display Decoder:**

* BCD is 4 bits, [0000 (0) – 1001 (9)] → Valid BCD's
  [1010 (10) – 1111 (15)] → Invalid BCD.
* 10 in BCD as,

  0001 0000

* The total number of segments to display BCD numbers from 0 to 9 is 7 segments [a, b, c, d, e, f, g].
* The seven segments are given below,

→ seven segment display

Digit	Segments Activated	Display
0	a, b, c, d, e, f	

1	b, c	b c
2	a, b, d, e, g	a g b e d
3	a, b, c, d, g	a g b c d
4	b, c, f, g	f b g c
5	a, c, d, f, g	a f g c d
6	a, c, d, e, f, g	a f g e c d
7	a, b, c	a b c
8	a, b, c, d, e, f, g	a f g b e c d
9	a, b, c, d, f, g	a f b g c d
10 - 15	x x x x x x x	Don't care conditions.

Digit	A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	0	1
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	1	1	1	1
7	0	1	1	1	1	1	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	1	0	1	1
10	1	0	1	0	X	X	X	X	X	X	X
11	1	0	1	1	X	X	X	X	X	X	X
12	1	1	0	0	X	X	X	X	X	X	X
13	1	1	0	1	X	X	X	X	X	X	X
14	1	1	1	0	X	X	X	X	X	X	X
15	1	1	1	1	X	X	X	X	X	X	X

Table: Truth table ~~for~~ BCD to 7-Segment decoder

For a

AB \ CD	00	01	11	10
00	1	0	1	1
01		1	1	1
11	X	X	X	X
10	1	1	X	X

$\boxed{a = A + C + BD + \bar{B}\bar{D}}$

For b

AB \ CD	00	01	11	10
00	1	1	1	1
01	1		1	
11	X	X	X	X
10	1	1	X	X

$\boxed{b = \bar{B} + \bar{C}\bar{D} + CD}$

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For c

AB \ CD	00	01	11	10
00	1	1	1	
01	1	1	1	1
11	X	X	X	X
10	1	1	X	X

$$\boxed{c = B + \bar{C} + D}$$

For d

AB \ CD	00	01	11	10
00	1		1	1
01		1		1
11	X	X	X	X
10	1	1	X	X

$$\boxed{d = \bar{B}\bar{D} + C\bar{D} + B\bar{C}D + \bar{B}C + A}$$

For e

AB \ CD	00	01	11	10
00	1			1
01				1
11	X	X	X	X
10	1		X	X

$$\boxed{e = \bar{B}\bar{D} + C\bar{D}}$$

For f

AB \ CD	00	01	11	10
00	1			
01	1	1		1
11	X	X	X	X
10	1	1	X	X

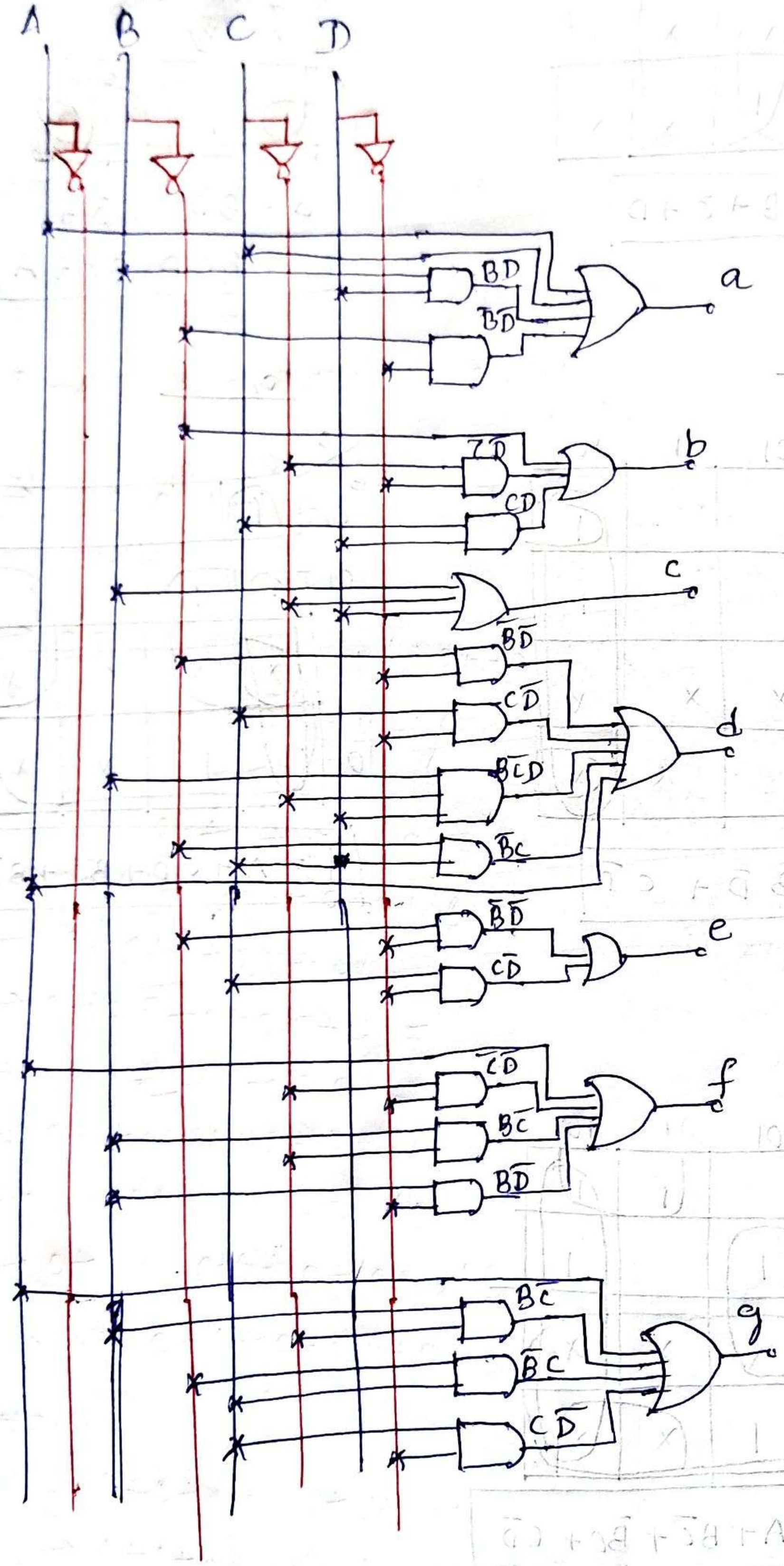
$$\boxed{f = A + \bar{C}\bar{D} + B\bar{C} + B\bar{D}}$$

For g:

AB \ CD	00	01	11	10
00			1	1
01	1	1		1
11	X	X	X	X
10	1	1	X	X

$$\boxed{g = A + B\bar{C} + \bar{B}C + C\bar{D}}$$

Logic diagram :



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